WORKED SOLUTION

STEP III 2013, Question 4

Pure • Roots of Unity Products

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s3-q4

This question has a full guided treatment in *Beyond STEP*, Section 5.3, *Factorising* $z^{2n} + 1$: https://beyondstep.keithmcnulty.org/ch05/z2n-plus-one.html

The key idea

The roots of $z^{2n} = -1$ come in conjugate pairs $e^{\pm i\theta}$, and each pair multiplies to a real quadratic, which gives the factorisation. Then a single well-chosen value of z turns the factorisation into a statement about cosines: at $z = i$ the $z^2 + 1$ part of every quadratic vanishes, leaving just the $-2z\cos\theta$ term. In part (ii) that same substitution would make both sides zero, so we first cancel the one factor that vanishes at $z = i$.

Solution

The factorisation

Expanding,

$$\begin{aligned}(z - e^{i\theta})(z - e^{-i\theta}) &= z^2 - z\left(e^{i\theta} + e^{-i\theta}\right) + e^{i\theta}e^{-i\theta} \\ &= z^2 - 2z\cos\theta + 1,\end{aligned}$$

since $e^{i\theta} + e^{-i\theta} = 2\cos\theta$.

Now $e^{i\theta}$ satisfies $z^{2n} = -1 = e^{i\pi}$ exactly when $2n\theta$ is an odd multiple of π, that is, $\theta = \frac{(2k-1)\pi}{2n}$ for an integer k. The values in $(-\pi, \pi]$ are

$$\theta = \pm\frac{\pi}{2n},\ \pm\frac{3\pi}{2n},\ \ldots,\ \pm\frac{(2n-1)\pi}{2n},$$

because $\frac{(2n-1)\pi}{2n} < \pi$ while $\frac{(2n+1)\pi}{2n} > \pi$. These are $2n$ distinct roots, so they are all the $(2n)$th roots of -1:

$$e^{\pm i\theta_k}, \qquad \theta_k = \frac{(2k-1)\pi}{2n}, \quad k = 1, 2, \ldots, n.$$

Since $z^{2n}+1$ is monic of degree $2n$ with these $2n$ distinct roots,

$$z^{2n}+1=\prod_{k=1}^{n}(z-\mathrm{e}^{\mathrm{i}\theta_k})(z-\mathrm{e}^{-\mathrm{i}\theta_k})=\prod_{k=1}^{n}\left(z^2-2z\cos\theta_k+1\right),$$

using the first result on each conjugate pair. This is the required identity.

Part (i)

Put $z=\mathrm{i}$. Each factor becomes

$$\mathrm{i}^2-2\mathrm{i}\cos\theta_k+1=-2\mathrm{i}\cos\theta_k,$$

and the left-hand side is $\mathrm{i}^{2n}+1=(-1)^n+1=2$, as n is even. So

$$2=(-2\mathrm{i})^n\prod_{k=1}^{n}\cos\theta_k.$$

With n even, $(-2)^n=2^n$ and $\mathrm{i}^n=(\mathrm{i}^2)^{n/2}=(-1)^{\frac{1}{2}n}$, so $(-2\mathrm{i})^n=2^n(-1)^{\frac{1}{2}n}$. Therefore

$$\prod_{k=1}^{n}\cos\theta_k=\frac{2}{2^n(-1)^{\frac{1}{2}n}}=(-1)^{\frac{1}{2}n}\,2^{1-n},$$

using $1/(-1)^{\frac{1}{2}n}=(-1)^{\frac{1}{2}n}$. The product on the left is exactly $\cos\frac{\pi}{2n}\cos\frac{3\pi}{2n}\cdots\cos\frac{(2n-1)\pi}{2n}$, as required.

Part (ii)

? **Ask yourself** Why not just put $z=\mathrm{i}$ in the factorisation again?

When n is odd, $\mathrm{i}^{2n}+1=(-1)^n+1=0$, so the left-hand side vanishes. On the right, the middle value $k=m$, where $m=\frac{1}{2}(n+1)$, gives $\theta_m=\frac{n\pi}{2n}=\frac{\pi}{2}$, so that factor is z^2+1, which is also zero at $z=\mathrm{i}$. We get $0=0$ and learn nothing. The cure is to remove the factor z^2+1 from both sides first.

Using the given fact and the factorisation,

$$(1+z^2)\left(1-z^2+z^4-\cdots+z^{2n-2}\right)=(z^2+1)\prod_{k\neq m}\left(z^2-2z\cos\theta_k+1\right).$$

These are polynomials, and z^2+1 is not the zero polynomial, so we may cancel it (polynomial division is unique):

$$1-z^2+z^4-\cdots+z^{2n-2}=\prod_{k\neq m}\left(z^2-2z\cos\theta_k+1\right),$$

an identity of polynomials, valid for every z, including $z=\mathrm{i}$.

Now substitute $z = \mathrm{i}$. On the left, the term $(-1)^j z^{2j}$ becomes $(-1)^j(-1)^j = 1$, and there are n terms ($j = 0, 1, \ldots, n-1$), so the left-hand side is n. On the right there are $n-1$ factors, each equal to $-2\mathrm{i}\cos\theta_k$:

$$n = (-2\mathrm{i})^{n-1} \prod_{k \neq m} \cos\theta_k.$$

As $n-1$ is even, $(-2\mathrm{i})^{n-1} = 2^{n-1}(-1)^{\frac{1}{2}(n-1)}$, exactly as in part (i).

To turn the product into squares, pair k with $n+1-k$. Since

$$\theta_{n+1-k} = \frac{(2n+1-2k)\pi}{2n} = \pi - \theta_k,$$

we have $\cos\theta_{n+1-k} = -\cos\theta_k$. The values $k \neq m$ split into the $\frac{1}{2}(n-1)$ pairs $\{k, n+1-k\}$ with $1 \leq k \leq \frac{1}{2}(n-1)$, so

$$\prod_{k \neq m} \cos\theta_k = \prod_{k=1}^{(n-1)/2} \left(-\cos^2\theta_k\right) = (-1)^{\frac{1}{2}(n-1)}\, P, \qquad P = \prod_{k=1}^{(n-1)/2} \cos^2\theta_k.$$

Putting this together, the two signs $(-1)^{\frac{1}{2}(n-1)}$ multiply to $+1$:

$$n = 2^{n-1}(-1)^{\frac{1}{2}(n-1)}(-1)^{\frac{1}{2}(n-1)} P = 2^{n-1} P,$$

so $P = n\,2^{1-n}$. The last factor in P has $k = \frac{1}{2}(n-1)$, that is, $\theta_k = \frac{(n-2)\pi}{2n}$, so P is precisely

$$\cos^2\frac{\pi}{2n}\cos^2\frac{3\pi}{2n}\cdots\cos^2\frac{(n-2)\pi}{2n} = n\,2^{1-n},$$

as required. (For $n = 1$ the product is empty, equal to $1 = 1 \cdot 2^0$, which is consistent.)

Checks

- $n = 2$ in part (i): $\cos\frac{\pi}{4}\cos\frac{3\pi}{4} = -\frac{1}{2} = (-1)^1 2^{-1}$. For $n = 4$ the product is $\frac{1}{8}$, matching $(-1)^2 2^{-3}$.
- $n = 3$ in part (ii): $\cos^2\frac{\pi}{6} = \frac{3}{4} = 3 \cdot 2^{-2}$. For $n = 5$, $\cos^2\frac{\pi}{10}\cos^2\frac{3\pi}{10} = 0.3125 = \frac{5}{16}$.
- Both identities were also confirmed by computer for all n up to 39, and the factorisation was tested at random complex z.

Common slips

- Substituting $z = \mathrm{i}$ straight into the factorisation in part (ii), which only gives $0 = 0$.
- Losing the sign: in part (i) the answer really is negative when $n \equiv 2 \pmod 4$, because $\cos\theta_k < 0$ for the $\frac{1}{2}n$ angles beyond $\frac{\pi}{2}$.
- Forgetting that $\cos\theta_{n+1-k} = -\cos\theta_k$, and so getting the wrong sign when forming squares.

Beyond STEP

The same trick, factorising $z^N \pm 1$ over its roots of unity and then evaluating at a clever point, gives a whole family of product formulas, such as $\prod_{k=1}^{n-1} \sin \frac{k\pi}{n} = \frac{n}{2^{n-1}}$ (try $z = 1$ in $\frac{z^n-1}{z-1}$). Letting $n \to \infty$ in such finite products leads to Euler's infinite product $\sin \pi x = \pi x \prod_{k \geq 1} \left(1 - \frac{x^2}{k^2}\right)$, a first taste of how complex analysis builds functions from their zeros. Needham's *Visual Complex Analysis* is a beautiful place to see roots of unity and these ideas geometrically.

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